Convolutional Networks for Plant Leaf Disease Detection: A Synthetic Sample Paper.

Abstract: This synthetic paper was created to demonstrate ResearchPilot AI with realistic content. We ask whether transfer learning with convolutional networks can detect plant diseases from leaf photographs collected by farmers.

Methodology: We fine tune a ResNet-18 convolutional network pretrained on ImageNet using 5000 labelled leaf photographs from the PlantVillage dataset, covering ten crop species and twelve disease categories. Images are resized to 224 by 224 pixels and augmented with random horizontal flips and slight colour jitter during training.

Findings: The fine tuned model achieves a macro F1 score of 0.91 on a held out test set, and detects the most frequent diseases with over 95 percent precision. Inference runs fast enough for mobile use. Limitations: All photographs were taken in laboratory conditions with uniform backgrounds, and the model has not been tested on field images where lighting and background vary.